

CSE-214 Online - 2 (B2)

Structural Design Pattern

Duration: 35 minutes

A retail shop originally sold only **individual food items** (e.g., burgers, pizzas, sandwiches, fries). Consequently, a customer order was simply maintained as a collection of food items.

As the business expanded, the shop introduced two new product categories:

- **Grocery items** (e.g., rice, flour, cooking oil, eggs), which are sold individually or as bundled packages. These packages contain individual items or other packages.
- **Set Menus**, which are predefined meal combinations consisting of one or more individual food items. A set menu **cannot** contain another set menu or any grocery items. The price of a set menu will be 10 percent less than the total price of the individual items.

A customer order can now include any combination of:

- Individual food items,
- Individual grocery items,
- One or more set menus,
- One or more grocery packages.

During checkout, the system should process all purchased items uniformly, regardless of whether they are individual items, set menus, or nested grocery packages.

Your Task

Implement the system should support the following operations:

1. Display the complete contents of an order in a clear, hierarchical format.
2. Calculate the total price of the order

The order class needs to be kept simple with a simple list of items all treated in the same way. This class is provided for your convenience. Your implementation should include this. Also a main is given with expected output. You have to code accordingly.

Your design should be flexible enough to accommodate future product categories and package types without requiring significant modifications to the existing implementation.

```
import java.util.ArrayList;
import java.util.List;

public class Order {

    private List<OrderItem> items = new ArrayList<>();

    public void add(OrderItem item) {
        items.add(item);
    }

    public double getTotalPrice() {

        double total = 0;

        for (OrderItem item : items) {
            total += item.getPrice();
        }

        return total;
    }

    public void printReceipt() {

        System.out.println("===== RECEIPT =====");

        for (OrderItem item : items) {
            item.print("");
        }

        System.out.println("-----");
        System.out.printf("Total Bill: £%.2f%n", getTotalPrice());
    }
}

public class Main {

    public static void main(String[] args) {

        // Foods
        Food burger = new Food("Burger", 8);
        Food pizza = new Food("Pizza", 10);
        Food fries = new Food("French Fries", 3);

        // Set Menu
        SetMenu lunch = new SetMenu("Lunch Combo");
```

```
lunch.addFood(burger);
lunch.addFood(fries);

// Grocery Items
Grocery rice = new Grocery("Rice", 20);
Grocery oil = new Grocery("Cooking Oil", 12);
Grocery eggs = new Grocery("Eggs", 6);
Grocery sugar = new Grocery("Sugar", 5);

// Small Package
GroceryPackage breakfastPack = new GroceryPackage("Breakfast Pack");
breakfastPack.add(eggs);
breakfastPack.add(sugar);

// Large Package (contains another package)
GroceryPackage monthlyPack = new GroceryPackage("Monthly Essentials");
monthlyPack.add(rice);
monthlyPack.add(oil);
monthlyPack.add(breakfastPack);

// Customer Order
Order order = new Order();

order.add(pizza);
order.add(lunch);
order.add(rice);
order.add(monthlyPack);

order.printReceipt();
}
```

===== RECEIPT =====

Food: Pizza (£10.00)

Set Menu: Lunch Combo

Food: Burger (£8.00)

Food: French Fries (£3.00)

Grocery: Rice (£20.00)

Package: Monthly Essentials

Grocery: Rice (£20.00)

Grocery: Cooking Oil (£12.00)

Package: Breakfast Pack
Grocery: Eggs (£6.00)
Grocery: Sugar (£5.00)

Total Bill: £84.00